

MATHEMATICS-I

Differential Calculus (Functions of one Variable):

Rolle's theorem

Cauchy's mean value theorem (Lagrange's mean value theorem as a special case) Taylor's and Maclaurin's theorems with remainders Indeterminate forms, Concavity and convexity of a curve, Points of inflexion, Asymptotes and Curvature.

Differential Calculus (Functions of several variables):

Limit, continuity and differentiability of functions of several variables Partial derivatives and their geometrical interpretation, differentials, derivatives of composite and implicit functions, derivatives of higher order and their commutativity Euler's theorem on homogeneous functions, harmonic functions Taylor's expansion of functions of several variables Maxima and minima of functions of several variables – Lagrange's method of multipliers.

Ordinary Differential Equations:

First order differential equations - exact, linear and Bernoulli's form Second order differential equations with constant coefficients Method of variation of parameters General linear differential equations with constant coefficients Euler's equations System of differential equations.

Sequences and Series :

Sequences and their limits Convergence of series Comparison test, Ratio test, Root test, Absolute and conditional convergence, alternating series Power series.

Complex Variables:

Limit, continuity, differentiability and analyticity of functions Cauchy-Riemann equations, line integrals in complex plane, Cauchy's integral theorem Independence of path, existence of indefinite integral, Cauchy's integral formula, derivatives of analytic functions, Taylor's series, Laurent's series, Zeros and singularities, Residue theorem, evaluation of real integrals.

MATHEMATICS-II

Linear Algebra: Algebra of matrices. Vector spaces - linear dependence of vectors, basis, linear transformations, rank and inverse of a matrix, solution of algebraic equations - consistency conditions, Hermitian, skew Hermitian and unitary matrices, bilinear forms, eigenvalues and eigenvectors. Numerical solution of system of linear equations à Gauss, Gauss-Jordan elimination and Gauss-Seidel iteration methods.

Integral Calculus: Fundamental theorem of integral calculus, mean value theorems, evaluation of definite integrals - reduction formulae. Convergence of improper integrals, tests of convergence, Beta and Gamma functions – elementary properties. Differentiation under integral sign, differentiation of integrals with variable limits - Leibnitz rule. Rectification, double and triple integrals, computations of area, surfaces and volumes, change of variables in double integrals - Jacobians of

transformations, integrals dependent on parameters - applications.

Vector Calculus: Scalar and vector fields, level surfaces, directional derivative, Gradient, Curl, Divergence, Laplacian, line and surface integrals, theorems of Green, Gauss and Stokes, line integrals independent of path. Numerical Analysis: Finite differences, Newton's forward and backward interpolation formulae, central difference interpolation formulae. Trapezoidal and Simpson's 1/3rd rules for numerical integration. Solution of polynomial and transcendental equations - bisection, Newton-Raphson and regula-falsi methods.

PROGRAMMING AND DATA STRUCTURE

Introduction to digital computers; introduction to programming à variables, assignments; expressions; input/output; conditionals and branching; iteration; functions; recursion; arrays; introduction to pointers; character strings; time and space requirements; searching and sorting; structures; introduction to data-procedure encapsulation; dynamic allocation; linked structures; introduction to data structures à stacks and queues. (A programming language like C/C++ may be used as a basis language. The same language must be used for the laboratory).

DESIGN & ANALYSIS OF ALGORITHMS

Prerequisite: Programming and Data Structures Basic concepts of algorithms, Complexity, Asymptotic notations, Trees: Binary tree, Binary Search Tree, Tree traversals. Heap as data structure. Basic sorting algorithms: selection sort, insertion sort. Greedy algorithms: Coin change problem, activity selection, Minimum Spanning Tree, Single source shortest path, knapsack problem. Divide – and – Conquer technique: Merge sort, quick sort. Solving Recurrence relations. Dynamic programming: matrix chain multiplication, all pair shortest path algorithm, 0-1 Knapsack. Graph algorithms: Warshall's algorithm, Depth First Search, Breadth First Search. Height Balanced Tree. Branch and Bound technique, Backtracking. NP completeness.

TRANSFORM CALCULUS

Laplace Transform : Definition of Laplace Transform, linearity property, conditions for existence of Laplace Transform. First and second shifting properties, Laplace Transform of derivatives and integrals, unit step functions, Dirac delta-function, error function. Differentiation and integration of transforms, convolution theorem, inversion, periodic functions. Evaluation of integrals by Laplace Transform. Solution of initial and boundary value problems. Fourier Series : Periodic functions, Fourier series representation of a function, half range series, sine and cosine series, Fourier integral formula, Parseval's identity. Fourier

Transform: Fourier Transform, Fourier sine and cosine transforms. Linearity, scaling, frequency shifting and time shifting properties. Self reciprocity of Fourier Transform, convolution theorem. Applications to boundary value problems. Brief Introduction of Z-Transform, Mellin transform and Wavelet Transform.

PARTIAL DIFFERENTIAL EQUATIONS

Review of power series solution of ODE, Frobenius series, Bessel functions and Legendre polynomials. Introduction to partial differential equations, linear and quasi-linear equations of first order. Classification of integrals. Lagrange's Method of solution and its geometrical interpretation, compatibility condition, Charpit's method, special types of first order equations. Second order partial differential equations with constant and variable coefficients, classification and reduction of second order equation to canonical form., characteristics. Cauchy problem, Cauchy's, Neumann and Dirichlet problems. Fourier series solution of wave equation, vibrations of a string. Riemann's method for hyperbolic equation. Method of separation of variables to solve heat equation, Laplace equation, Diffusion equation. Integral transform method to solve second order partial differential equations.

PROBABILITY & STATISTICS

Probability: Classical, relative frequency and axiomatic definitions of probability, addition rule and conditional probability, multiplication rule, total probability, Bayes' Theorem and independence. Random Variables: Discrete, continuous and mixed random variables, probability mass, probability density and cumulative distribution functions, mathematical expectation, moments, moment generating function, Chebyshev's inequality. Special Distributions: Discrete uniform, Binomial, Geometric, Poisson, Exponential, Gamma, Normal distributions. Functions of a Random Variable. Joint Distributions: Joint, marginal and conditional distributions, product moments, correlation, independence of random variables, bivariate normal distribution. Sampling Distributions: The Central Limit Theorem, distributions of the sample mean and the sample variance for a normal population, Chi-Square, t and F distributions. Estimation: The method of moments and the method of maximum likelihood estimation, confidence intervals for the mean(s) and variance(s) of normal populations. Testing of Hypotheses: Null and alternative hypotheses, the critical and acceptance regions, two types of error, power of the test, the most powerful test and Neyman-Pearson Fundamental Lemma, tests for one sample problems for normal populations.

NUMERICAL SOLUTION OF ORDINARY AND PDE

Ordinary Differential Equations: Numerical solutions of IVP

- Difference equations, stability, error and convergence analysis. Single step methods - Taylor series method, Euler method, Picard's method of successive approximation, Runge Kutta Method. Multi step methods - Predictor-Corrector method, Euler PC method, Milne and Adams Moulton PC method. System of first order ODE, higher order IVPs. Numerical solutions of BVP - Linear BVP, finite difference methods, shooting methods, Newton's method for system of equations, stability, error and convergence analysis, non linear BVP, higher order BVP. Partial Differential Equations: Classification of PDEs, Finite difference approximations to partial derivatives, convergence and stability analysis. Explicit and Implicit schemes - Crank-Nicolson scheme, tri-diagonal system, Laplace equation using standard five point formula and diagonal five point formula. ADI scheme, hyperbolic equation, explicit scheme, method of characteristics. Solution of one dimensional heat conduction equation by Schmidt and Crank Nicolson methods. Solution of wave equation.

DISCRETE MATHEMATICS

Cartesian product, relations, domain and range, composition of relations, equivalence relations, partial ordering relation. Lattices. Functions -Function as a relation, injection, surjection and bijection, composition of functions, identity and inverse function. Cardinality, characteristic function. Peano postulates and finite induction, example of proof by induction. Recursive definitions. Binary operation on a set, groupoid, commutative and associative binary operations, binary operation with identity. Semigroup, monoid. Boolean algebra - Axioms and properties, atomic structure of a finite Boolean algebra. Homomorphism and isomorphism. Disjunctive and conjunctive normal forms. Algebra of position. First order predicate calculus. Theory of groups - Axioms, properties, subgroup, cyclic group, cosets, Lagrange's theorem. Rings, subrings, ideals, ring homomorphism. Finite field - Field of order p. (p is prime).

SOFT- COMPUTING TOOLS IN ENGINEERING

Fuzzy Logic, Genetic Algorithm, Neural Networks and Probabilistic Reasoning; Applications of Fuzzy Logic concepts in Engineering Problems; Engineering optimization problem solving using genetic algorithm; Neural network approaches in engineering analysis, design and diagnostics problems; applications of probabilistic reasoning approaches.

OBJECT ORIENTED SYSTEMS DESIGN

Prerequisite: Programming and Data Structures
Objects, Classes, object modeling, object relationships; Static and dynamic modeling, object oriented language features; JAVA, UML; Object oriented analysis and design,

client server and distributed objects, client server implementation; CORBA, COM/DCOM.

COMPUTER ORGANISATION & ARCHITECTURE

Computer arithmetic, point representations, introduction to CISC processor architecture, instruction set and addressing modes. Hardware design principles, polling of processors, memory types and interfacing and timing I/O handling, interrupts and DMA and device interfaces-CRT, floppy disk, HDD, optical disk, serial interfaces and data acquisition, operating system concepts and architectural support- privileged mode, software interrupts, memory hierarchy and virtual memory, multi-processors concept, cache memory, pipe lining and introduction, RISC processors, super scalar processors. Resource scheduling problems-their solutions and analysis, storage management, secondary storage, virtual memory, segmentation and paging. Concurrent processes and their synchronization abstractions. Producer & consumer problems, mutual exclusion and dead-lock, properties of concurrent programs, critical section problem, semaphores, test and set, atomic operations, security and recovery.

MATHEMATICAL METHODS

Two-point boundary-value problems, Green's functions, Construction of Green's functions, Nonhomogeneous boundary conditions, Sturm- Liouville Systems, Eigen values and Eigen functions, Eigenfunction expansions and completeness.; Hypergeometric equation and functions, Properties of hypergeometric functions, Legendre equation and Legendre polynomial, Generating function for Legendre polynomial, Recurrence relations between Legendre polynomials,Rodrigue's formula. Orthogonality of Legendre polynomial, Associated Legendre equation and Legendre function, Bessel equation and its solution, Bessel functions, Modified Bessel function, Generating function for Bessel function, Recurrence relations between Bessel functions, Orthogonality of Bessel functions.; Autonomous systems, Stability for Linear systems with constant coefficients, Linear plane autonomous systems, perturbed systems, Method of Lyapunov for nonlinear systems. Limit cycles of Poincare.; Coordinate transformations, Definition of Tensors, Summation convention, Kronecker Delta, Covariant, contravariant and mixed tensors. Fundamental operations with tensors, the line element and metric tensor, length of a vector, Christoffel's symbols, the covariant derivative, tensor form of gradient, divergence and curl. Examples from continuum mechanics, elasticity, plasticity, fluids

LINEAR ALGEBRA

Vector spaces over any arbitrary field, linear combination, linear dependence and independence, basis and dimension, inner-product spaces, linear transformations,

matrix representation of linear transformations, linear functional, dual spaces, eigen values and eigen vectors, rank and nullity, inverse and linear transformation, Cayley-Hamilton Theorem, norms of vectors and matrices, transformation of matrices, adjoint of an operator, normal, unitary, hermitian and skew-hermitian operators, quadratic forms.

REAL ANALYSIS

Dedekind's definition of real numbers, field and order axioms, countable and uncountable sets, supremum and infimum of sets of real numbers, bounds and limit points of a set, Bolzano-Weierstrass theorem, open and closed sets. Limit inferior, limit superior and limit of sequence, bounded and monotonic sequences, Cauchy sequence and Cauchy's general principle of convergence, product and quotient of limits, Cantor's theorem on nested interval and its applications. Compact sets, Heine-Borel theorem. Limit, limit superior, limit inferior of real functions, limit theorems. Continuity and uniform continuity of real functions, properties of continuous functions, continuity and compactness. Differentiability of real functions, Taylor's and Maclaurin's theorems. Riemann integration, conditions for integrability, properties of integrable functions, indefinite integral and their properties, fundamental theorem of integral calculus, mean value theorems, improper integrals, convergence at infinity, absolute and conditional convergence. Sequences and series of functions, uniform convergence of sequences and series of functions. Cantor's definition of real numbers. Metric sets: Definition, real line as an example of a metric set.

OPERATION RESEARCH

Baseline model, linear programming problem, convex sets, convex functions and their properties, basic feasible solution, optimal solution, related theorems. Graphical method for solving two and three variable problems, simplex method, Big M method, degenerate LP problem, product form of inverse of a matrix, revised simplex method, duality theorems, complementary slackness principle, primal-dual simplex algorithm, sensitivity analysis, parametric programming, linear integer programming problem, Gomory cutting plane method, branch and bound algorithm, 0-1 implicit enumeration, transportation problem, assignment problem with their solution methodologies. Theory of games, two-person zero-sum games with and without saddle-points, pure and mixed strategies, graphical method of solution of a 2 x n game, solution of an m x n game by simplex method.

MODERN ALGEBRA

Groups, subgroups, cyclic groups, cosets, Lagrange theorem. Normal subgroup, quotient group; homomorphism, fundamental theorem of

homomorphism; permutation group, Cayley's theorem; direct product of group. Commutative ring with identity: Axioms, examples, integral domain; field, ideals, quotient ring, prime and maximal ideal, principal ideal domain; Euclidean domain, the field of quotients of an integral domain; polynomial ring over a field. Roots of polynomials, extension of fields, splitting fields.

MEASURE THEORY AND INTEGRATION

Prerequisite: Real Analysis

Algebra of sets, ring, sigma-ring, field and sigma field of sets, monotone class, Lebesgue measure and outer measure, measurable sets, measurable functions, Littlewoods three principles, existence of non-measurable set. Lebesgue integral of a bounded function over a set of finite measure, the integral of a non-negative function, general Lebesgue integral, convergence in measure, functions of bounded variation, absolute continuity, differentiation and integration, general measure and integration, signed measure, Hahn-Jordan decomposition, Radon-Nikodym and Lebesgue decomposition theorems, product measures and Fubini's theorem. L_p spaces, Minkowski and Holder inequalities, convergence and completeness approximation in L_p , bounded linear functionals on L_p spaces.

SWITCHING & FINITE AUTOMATA

Codes, switching algebra, minimization and decomposition of switching functions, Karnaugh map, digital circuits, logical gates - AND, OR, NAND, NOR etc., combinatorial logic, threshold logic, fault diagnosis, sequential machines - synchronous and asynchronous circuits, basic flip-flops - SR flip-flop, T flip-flop, JK flip-flop, flip-flop, clocked and master-slave flip-flop, decoder, multiplexer circuits, counters - synchronous and asynchronous, sequence detector, parity-bit generator, registers, shift registers, minimization and transformation of sequential machines, Turing machines.

ADVANCED NUMERICAL TECHNIQUES

Revision of IVP, single-step and multi-step methods. System of first order ODE, higher order IVPs. Numerical solutions of BVP - Linear BVP, finite difference methods, shooting methods, stability, error and convergence analysis, non linear BVP, higher order BVP. Classification of PDEs, Finite difference approximations to partial derivatives, convergence and stability analysis. Explicit and Implicit schemes - Crank-Nicolson scheme, tri-diagonal system, Laplace equation using standard five point formula and diagonal five point formula. ADI scheme, hyperbolic equation, explicit scheme, method of characteristics. Solution of one dimensional heat conduction equation by Schmidt and Crank Nicolson methods. Solution of wave equation.

FUNCTIONAL ANALYSIS

Metric spaces, complete metric spaces, Banach contraction mapping theorem. Banach spaces, bounded linear functionals and bounded linear operators, dual spaces, Hahn-Banach theorem, uniform boundedness principle, open mapping and closed graph theorems, weak convergence, Hilbert spaces, orthonormal sets, Riesz representation theorem, bounded linear operators on Hilbert spaces.

SYSTEMS PROGRAMMING

Prerequisite: Programming and Data Structures
Computer structure, data representation, instruction format, assembly language, addressing systems. Macrolanguage and macroprocessors, concepts and design of assemblers, loaders, linkage editors.

FLUID MECHANICS

Prerequisite: PDE Kinematics of Fluids in Motion : Continuum Hypothesis, Lagrangian and Eulerian description, Introduction to stream lines, velocity potential, vorticity vector etc., Equation of continuity. Equations of Motion, Euler's equations of motion, Bernoulli's equation. Potential flows. Three-dimensional flows : Singularities and image systems. Weiss sphere theorem, axi-symmetric flows, Stokes stream function. Two-dimensional flows : stream function and complex potential for two-dimensional, irrotational incompressible flows, two-dimensional image systems, Milne-Thomson circle theorem and its applications, Blasius theorem, use of conformal transformations, Kutta-Joukowski condition, Karman vortex street. Viscous flows : stress analysis in fluid motion, relations between stress and rate of strain, Navier-Stokes equations of motion of a viscous fluid, some exact solutions of Navier-Stokes equations, flow past a sphere, Prandtl's boundary layer theory, Karman's integral equation, inviscid compressible flow - Propagation of pressure change.

NUMBER THEORY

Prerequisite: Modern Algebra, Real Analysis
Peano's axioms, divisibility, properties of integers and prime numbers, fundamental theorem of arithmetic. Congruences, solutions of congruences, congruences of degree one, congruences of higher degree. Quadratic residues, quadratic reciprocity, Jacobi symbol, greatest integer function, arithmetic functions, the Mobius inversion formula, multiplication of arithmetic functions, recurrence functions, some Diophantine equations, simple continued fractions, distribution of primes, algebraic numbers, algebraic number fields, partition function.

QUEUEING THEORY

Prerequisite: Prob. Theory/ Prob. and Stats. Probability and random variable, discrete and continuous univariate and multivariate distributions, moments, law of large numbers and central limit theorem (without proof), Poisson process, birth and death process, infinite and finite queueing models M/M/1, M/M/C, M/G/1, M/M/1/N, M/Ek/1, Ek/M/1, M/G/1/N, GI/M/1, priority queueing models, network of queues, finite processor sharing models, central server model of multiprogramming, performance evaluation of systems using queueing models. Concepts of bottleneck and system saturation point. Introduction to discrete time queues and its applications.

REGRESSION AND TIME SERIES MODEL

Concept of regression, Simple linear regression, multiple linear regression, model adequacy checking, transformations and weighting to correct model inadequacies, diagnostics for leverage and influence. Polynomial regression models, orthogonal polynomials. Classical techniques of Time Series Analysis, Different Smoothing Techniques, General linear process, Autoregressive Processes AR(P), Moving average Process Ma(q): Autocorrelation, Partial autocorrelation and Spectrum, Identification in time domain, Forecasting, Estimation of Parameters, Model diagnostic checks, Elements of ARCH and GARCH models, Use of time series techniques in Finance.

STOCHASTIC PROCESSES

Prerequisite: Probability and Statistics Definition and classification of stochastic processes, discrete time Markov chains, random walk, gamblers ruin, branching process, Poisson process, general continuous time Markov chains, birth and death process, applications to queues, renewal process, martingales, Brownian motion.

INTEGRAL EQUATIONS & VARIATIONAL METHODS

Integral Equations: Basic concepts, Volterra integral equations, relationship between linear differential equations and Volterra equations, resolvent kernel, method of successive approximations, convolution type equations, Volterra equation of the first kind. Abel integral equation. Fredholm integral equations, Fredholm equations of the second kind, the method of Fredholm determinants, iterated kernels, integral equations with degenerate kernels, eigen values and eigen functions of a Fredholm alternative, construction of Green's function for BVP, singular integralequations. Calculus of Variations: Euler - Lagrange equations, degenerate Euler equations, Natural boundary conditions, transversality conditions, simple applications of variational principle, sufficient conditions for extremum. Variational formulation of BVP, minimum of quadratic functional. Approximate methods - Galerkin's method, weighted-residual methods, Collocation

methods. Variational methods for time dependent problems.

THEORY OF OPERATING

Prerequisite: Programming and Data Structures CPU, memory, file and I/O devices, batch processing, resource scheduling problems - their solutions and analysis, storage management, secondary storage, virtual memory, segmentation and paging. Concurrent processes and their synchronization abstractions. Producer consumer problem, mutual exclusion and deadlock, properties of concurrent programmes, critical section problem, semaphors, test and set, atomic operations, security and recovery.

FILE ORGANISATION AND DATA BASE SYSTEMS

Prerequisite: Programming Languages Database concepts, entity relationship model, relational network and hierarchical data models. Relational database design, functional and multi-valued dependency. Normal form, data description languages, relational algebra and calculus, SQL, query facility and query optimization. Introduction to distributed databases implementation issues. Storage media and storage structure, file and index organization. Heap file, indexed file, B tree file, denie index, file with variable length records, performance and evaluation.

GRAPH THEORY AND ALGORITHMS

Graphs, paths and circuits, trees and fundamental circuits, cut-sets and cut-vertices, planar and dual graphs, colouring, covering and partitioning, direct graphs, enumeration of graphs, graph theoretic algorithms and applications.

COMPUTATIONAL STATISTICS

Prerequisite: Probability and Statistics Analysis of Variance, one-way and two-way classification, Concept of design of experiment. Some standard designs: completely randomized design, randomized block design, Latin Squares, Graeco Latin Squares, and factorial designs, confounding and blocking in factorial designs, fractional factorial designs. Simple and multiple regression models. Classical techniques of time series analysis, smoothing and decomposition. Analysis of covariance model.

SPEECH & NATURAL LANGUAGE PROCESSING

Major Components of the Course include:

1. Basic Text Processing: Tokenization, Stemming
2. Language Modeling: N-grams, smoothing
3. Morphology, Parts of Speech Tagging

4. Syntax: PCFGs, Dependency Parsing
5. Distributional Semantics
6. Lexical Semantics, Word Sense Disambiguation
7. Information Extraction: Relation extraction, Event extraction
8. Text Summarization
9. Text Classification
10. Machine Translation

TOPOLOGY

Prerequisite: Real Analysis

Topological spaces, weak topology, subspace topology, product and quotient spaces, continuous maps and homeomorphism, Hausdorff spaces, compact and locally compact spaces, separation axioms, connectedness, paths, equivalence classes of paths, path connected spaces.

NON LINEAR PROGRAMMING

Prerequisite: Operations Research
 Convex set, Convex function, Generalized convex functions. Fritz John and Karush-Kuhn-Tucker optimality condition, duality, Convex programming problems, Quadratic programming, Fractional programming, Separable programming, Non linear integer programming. Constrained Optimization: One dimensional search methods, Multi-dimensional search methods. Unconstrained optimization: Conjugate gradient method, Generalized reduced gradient methods, Method of feasible direction.

COMPUTATIONAL LINEAR ALGEBRA

Basic concepts, floating point numbers and errors in computation, stability of algorithms and conditioning of problems. Numerical solution of linear system, Direct methods : Gaussian elimination with pivotal condensation, operational count and error bound. LU factorization, QR factorization. Condition number and ill conditioned systems. Matrix and vector norms. Error bounds. Wilkinson's algorithm for ill-conditioned systems. Iterative methods : Jacobi, Gauss-Seidel, SOR, convergence and rate of convergence, conjugate gradient method, Arnoldi process and GMRES. Large sparse systems. Matrix inverse. Generalized inverse. Least square solution of linear systems. Numerical eigenvalue problems. computation of eigenvalues and eigenvectors, singular value decomposition and least square problem, SVD and the pseudo inverse. Jacobi, Givens & Householder's method for symmetric matrices, Hessenberg QR iteration